

GRAND MANAN, NB, Canada

WMO: 710300

Lat: 44.712N

Lon: 66.802W

Elev: 257

StdP: 14.56

Time Zone: -4.00 (NAA)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.5	4.1	-9.8	3.3	2.7	-5.7	4.1	7.5	28.4	32.9	24.6	32.0	10.5	300	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.3	79.1	67.1	76.2	65.2	73.6	64.0	69.7	76.2	67.7	73.2	66.0	70.8	7.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	101.6	72.6	65.7	95.7	70.3	64.2	90.8	68.3	33.9	76.6	32.3	73.1	30.9	70.9	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	19.6	17.4	-6.5	85.5	4.3	2.7	-9.5	87.4	-12.0	89.0	-14.4	90.5	-17.5	92.4		
(5)			-6.9	73.5	4.3	2.0	-9.9	75.0	-12.5	76.2	-14.9	77.3	-18.0	78.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	23.2	24.2	30.6	40.0	48.9	57.2	63.7	63.5	57.2	47.6	39.0	29.6		
	DBStd	15.87	10.37	9.15	8.14	5.89	5.49	5.55	4.46	4.37	6.36	6.72	7.69	9.29		
	HDD50	3663	832	722	603	302	87	7	0	0	12	129	337	632		
	HDD65	7811	1297	1142	1067	749	500	242	77	78	243	540	779	1097		
	CDD50	1412	0	0	1	3	53	223	424	420	227	54	7	0		
	CDD65	85	0	0	0	0	0	8	36	33	8	0	0	0		
	CDH74	524	0	0	1	0	15	72	214	176	44	2	0	0		
	CDH80	67	0	0	0	0	2	8	29	21	7	0	0	0		
	WSAvg	8.2	9.5	9.9	9.8	8.7	8.1	7.2	6.3	6.1	6.8	8.1	8.8	9.6		
	PrecAvg	50.1	4.5	3.7	4.4	3.7	4.0	3.7	3.2	3.2	4.0	5.0	4.8	5.3		
(15) Precipitation	PrecMax	70.9	8.3	7.5	7.0	8.1	8.2	10.1	7.6	9.2	8.5	13.1	7.7	12.5		
	PrecMin	29.3	1.9	1.1	0.9	0.5	1.2	1.3	0.7	0.6	1.5	1.5	2.7	1.9		
	PrecStd	8.3	1.5	1.4	1.6	1.9	1.7	2.1	1.9	1.9	2.0	2.7	1.5	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.9	48.2	56.0	62.7	75.8	81.0	83.8	83.2	78.8	70.6	58.8	53.5		
		MCWB	50.2	46.3	45.9	51.1	60.8	67.1	70.3	70.9	68.0	59.6	56.3	52.8		
	2%	DB	46.7	43.7	48.1	56.5	67.9	75.5	79.0	78.1	73.8	63.6	56.5	50.4		
		MCWB	45.9	41.6	42.3	47.1	56.5	63.4	67.5	66.4	64.2	58.3	54.9	49.6		
	5%	DB	42.8	40.4	44.3	52.7	62.9	71.5	76.1	75.5	69.9	60.5	54.3	47.2		
		MCWB	41.2	38.5	41.0	45.5	53.2	60.7	65.7	65.2	62.2	57.1	52.6	45.7		
	10%	DB	38.6	37.8	41.3	49.6	58.7	68.0	73.2	72.8	66.8	58.1	51.6	43.9		
		MCWB	36.8	35.8	37.9	44.2	51.1	58.6	64.2	64.3	60.8	55.3	49.7	41.8		
	0.4%	WB	50.4	47.4	49.2	53.7	64.3	68.8	72.3	72.5	69.8	63.5	57.9	52.9		
		MCDB	50.8	47.8	53.2	58.2	73.1	77.7	79.6	80.4	76.1	65.9	58.2	53.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.9	42.5	44.6	50.3	58.3	65.9	69.7	69.8	66.3	60.3	55.6	49.7		
		MCDB	46.5	43.1	46.2	52.7	64.9	73.2	76.7	75.6	70.8	62.2	55.9	50.2		
	5%	WB	41.5	39.0	41.5	47.6	55.1	63.0	67.8	67.7	64.3	58.1	53.0	46.3		
		MCDB	42.6	40.1	43.7	50.9	60.2	68.6	73.5	72.3	67.6	59.8	53.9	47.0		
	10%	WB	36.8	36.2	38.4	45.2	52.7	60.5	65.9	65.9	62.6	55.8	50.0	42.3		
		MCDB	38.4	37.7	40.8	48.3	56.9	65.5	71.1	69.7	65.1	57.7	51.2	43.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	15.3	14.1	14.5	16.2	16.9	16.9	17.3	17.4	15.2	13.6	13.7		
		MCDBR	18.5	16.8	17.0	20.3	24.3	23.7	21.8	21.5	19.8	17.7	14.8	17.0		
		MCWBR	19.2	16.3	13.4	13.1	14.3	13.3	12.1	12.0	12.4	13.1	13.9	17.7		
		MCDBR	18.6	16.7	15.2	15.7	19.3	20.0	18.3	17.0	14.9	14.3	13.5	16.8		
	5% WB	MCWBR	19.5	16.9	14.0	12.2	13.2	13.0	11.6	11.1	10.7	12.7	14.2	17.8		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.301	0.328	0.353	0.372	0.407	0.424	0.400	0.364	0.335	0.317	0.300		
	taud		2.477	2.493	2.454	2.425	2.418	2.343	2.323	2.391	2.455	2.530	2.534	2.491		
	Ebn at Noon		262	284	288	289	285	274	267	270	272	267	253	249		
	Edh at Noon		22	26	31	35	36	39	39	36	31	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		447	732	1125	1413	1603	1757	1793	1600	1308	839	488	354		
	RadStd		38	61	81	136	193	157	140	102	89	83	49	34		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.77	N/A	N/A	+97	N/A	N/A

Nomenclature: See separate page